

ABHINAY MESHRAM

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SUMMARY

Machine Learning Engineer with hands-on experience building Generative AI and Machine Learning applications using Python, LangChain, Google Gemini, RAG, ChromaDB, FastAPI, Streamlit, Docker, and AWS EC2. Experienced in LLM applications, document retrieval pipelines, vector embeddings, conversational AI, REST APIs, model optimization, and end-to-end ML workflows.

SKILLS

Programming Languages: Python, SQL

GenAI & LLM: Generative AI, LLMs, LangChain, Google Gemini, Google GenAI SDK, Prompt Engineering, RAG, Conversational AI

RAG & Vector DB: ChromaDB, FAISS, Vector Embeddings, Semantic Search, Similarity Search, Document Processing, Text Splitting

ML/NLP: Scikit-learn, NLP, Text Preprocessing, Classification, Regression, Feature Engineering, Model Evaluation, SMOTE

Deep Learning: Artificial Neural Networks (ANN), Convolutional Neural Network (CNN)

Data Analysis & Visualization: Exploratory Data Analysis, Statistical Analysis, Data Cleaning, Matplotlib, Seaborn

Libraries & Frameworks: Pandas, NumPy, TensorFlow, Keras **Frameworks & Deployment:** FastAPI, Streamlit, Docker, Git, GitHub, AWS EC2

WORK EXPERIENCE

Machine Learning Intern

Jun 2026 – Jul 2026

Cognifyz IT Solutions Pvt. Ltd. — Remote

- Developed machine learning solutions for restaurant analytics by completing projects involving restaurant rating prediction, recommendation systems, and cuisine classification.
- Performed data preprocessing, handled missing values, encoded categorical features, and prepared datasets for machine learning model development.
- Built and evaluated machine learning models for regression, classification, and content-based recommendation tasks using Python and Scikit-learn.

PROJECTS

RAG Document Assistant (Python, LangChain, Google Gemini, ChromaDB, RAG, FastAPI, Docker) [Link](#) Aug 2026

- Built a RAG application for querying information from uploaded documents using LangChain and Google Gemini.
- Developed a retrieval pipeline covering document loading, text splitting, vector embeddings, ChromaDB storage, and similarity-based retrieval.
- Integrated retrieved context with LLM generation and structured the application for FastAPI API deployment and Docker containerization.

AI Persona Assistant (Python, Google Gemini, Google GenAI SDK, Streamlit) [Link](#)

Aug 2026

- Built a conversational AI application using Google Gemini with configurable Coding Tutor, Friendly Assistant, and Sarcastic Buddy personas.
- Implemented custom conversation memory to maintain chat history and provide context-aware responses.
- Developed a Streamlit chat interface with persona selection, persona switching, conversation management, and streaming LLM responses.

Credit Card Fraud Detection (Python, Scikit-learn, XGBoost, FastAPI, Streamlit) [Link](#)

Jun 2026

- Built an end-to-end fraud detection system on **284,807** credit card transactions, performing EDA, feature scaling, SMOTE oversampling, and model comparison using Scikit-learn and XGBoost.
- Optimized a tuned **XGBoost** model using **RandomizedSearchCV** and **Stratified K-Fold Cross-Validation**, achieving **ROC-AUC 0.977**, **PR-AUC 0.869**, **93% precision**, and **82% recall**.
- Developed a **FastAPI REST API** and integrated it with a **Streamlit** dashboard for real-time and batch fraud prediction, probability scoring, threshold tuning, and interactive visualizations.

CERTIFICATIONS

- Machine Learning Mastery
- Data Science & Analytics with Artificial Intelligence
- Data Analysis with Python

EDUCATION

Bachelor of Science in Information Technology

University of Mumbai, Mumbai, India

Jun 2022 – Apr 2025

CGPA: 8.22 / 10